

Student lifestyle

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Data collection & Exploration

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Data Acquisitions and Analysis The data ranges for an academic year from August 2023 up to May 2024, and thus a better understanding of the student lifestyles is based mainly from India. It has 100 rows (observations/records) and 8 columns or what can be referred to as features or variables. It offers details about student's events of mass, healthy and unhealthy habits and performance of students in mass daily activities which includes their learning, extra curricula activities, sleeping, social and physical activities. Stress is computed for each student through studying and sleeping hours to show how lifestyle factors may affect performance.

Columns:

1. Student ID (Categorical)

- A unique identifier assigned to each student, important for ensuring data accuracy and maintaining individual-level information.

2. Study Hours Per Day (Quantitative)

- The number of hours a student spends studying daily, which may correlate with academic performance.

3. Extracurricular Hours Per Day (Quantitative)

- The number of hours spent on extracurricular activities daily, providing insights into student involvement outside academics.

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4. Sleep Hours Per Day (Quantitative)

- The number of hours of sleep students get daily, crucial for understanding its potential impact on performance and well-being.

5. Social Hours Per Day (Quantitative)

- The daily hours spent on social interactions, which may reflect work-life balance or emotional well-being.

6. Physical Activity Hours Per Day (Quantitative)

- Hours spent on physical activities daily, highlighting the role of physical health in a student's lifestyle.

7. GPA (Quantitative)

- The Grade Point of the student, a key metric of academic success.

8. Stress Level (Categorical)

- Describes stress levels as Low, Moderate, or High, indicating mental health status and its potential link to other variables.

ID	Study_Hours_Per_Day	Extracurricular_Hours_Per_Day	Sleep_Hours_Per_Day	Social_Hours_Per_Day	Physical_Activity_Hours_Per_Day	GPA	Stress_Level
1	6.90	6.90	3.80	8.70	2.80	1.80	2.99Moderate
2	5.30	5.30	3.50	8.00	4.20	3.00	2.75Low
3	5.10	5.10	3.90	9.20	1.20	4.60	2.67Low
4	6.50	6.50	2.10	7.20	1.70	6.50	2.88Moderate
5	8.10	8.10	0.60	6.50	2.20	6.60	3.51High
6	6.00	6.00	2.10	8.00	0.30	7.60	2.85Moderate
7	8.00	8.00	0.70	5.30	5.70	4.30	3.08High
8	8.40	8.40	1.80	5.60	3.00	5.20	3.20High
9	5.20	5.20	3.60	6.30	4.00	4.90	2.82Low
10	7.70	7.70	0.70	9.80	4.50	1.30	2.76Moderate
11	9.70	9.70	3.60	8.00	2.50	0.20	3.43High
12	6.90	6.90	1.10	9.10	2.10	4.80	2.97Moderate
13	6.40	6.40	2.20	5.70	4.80	4.90	2.82High
14	5.00	5.00	3.30	8.50	4.40	2.80	2.87Low
15	8.90	8.90	0.30	6.80	0.70	7.30	3.40High
16	6.70	6.70	0.30	6.60	2.00	8.40	3.20Moderate
17	8.60	8.60	2.60	9.40	1.60	1.80	3.36High
18	8.80	8.80	2.20	8.90	2.00	2.10	3.19High
19	7.60	7.60	1.70	5.10	0.60	9.00	3.16High
20	6.60	6.60	2.00	9.50	1.50	4.40	2.93Moderate
21	7.10	7.10	3.00	6.10	0.50	7.30	2.90Moderate
22	9.60	9.60	3.20	8.20	2.60	0.40	3.34High
23	9.00	9.00	0.70	9.50	2.60	2.20	3.21High
24	6.10	6.10	1.70	9.10	5.20	1.90	2.85Moderate
25	5.00	5.00	2.00	7.10	1.30	8.60	2.68Low
26	9.70	9.70	1.30	7.60	3.80	1.60	3.29High
27	6.80	6.80	3.90	9.80	0.90	2.60	2.64Moderate
28	6.40	6.40	0.10	8.00	3.00	6.50	2.95Moderate
29	5.30	5.30	1.10	9.50	1.40	6.70	2.79Low
30	9.90	9.90	1.00	8.40	3.60	1.10	3.25High
31	6.20	6.20	2.90	6.80	3.80	4.30	3.05Moderate
32	5.50	5.50	3.30	6.60	1.10	7.50	3.26Low
33	5.20	5.20	2.40	8.40	0.10	7.90	2.47Low
34	8.20	8.20	0.70	8.50	2.30	4.30	3.24High
35	9.70	9.70	0.60	6.70	0.70	6.30	3.62High
36	9.10	9.10	2.20	7.60	1.20	3.90	3.12High
37	5.50	5.50	3.60	9.50	3.40	2.00	2.58Low
38	8.60	8.60	3.60	9.40	1.90	0.50	3.03High
39	8.20	8.20	0.30	5.80	5.40	4.30	3.29High
40	5.00	5.00	0.60	7.70	4.20	6.50	2.61Low
41	8.30	8.30	0.90	8.60	1.40	4.80	3.48High
42	8.20	8.20	3.40	8.30	2.30	1.80	3.07High
43	5.50	5.50	1.50	6.30	1.50	9.20	2.81Low
44	9.50	9.50	2.50	9.00	1.50	1.50	3.49High
45	7.90	7.90	2.00	6.00	4.30	3.80	2.97Moderate
46	9.70	9.70	3.80	9.60	0.30	0.60	3.55High
47	5.10	5.10	3.70	7.10	5.80	2.30	2.60Low
48	9.30	9.30	1.30	5.80	3.30	4.30	3.09High
49	9.70	9.70	2.80	7.90	0.30	3.30	3.47High
50	9.40	9.40	3.00	8.50	2.20	0.90	3.18High

The Dataset

51	6.80	1.20	9.00	4.90	2.10	3.12Moderate
52	9.00	2.60	8.50	3.10	0.80	4.00High
53	9.50	1.40	6.90	0.60	5.60	3.33High
54	7.30	2.20	6.40	3.50	4.60	3.11Moderate
55	5.20	0.10	9.10	2.20	7.40	2.79Low
56	8.80	0.90	8.10	0.50	5.70	3.10High
57	5.30	2.10	7.70	3.80	5.10	2.39Low
58	9.00	1.10	7.20	0.50	6.20	3.39High
59	5.10	3.90	9.20	4.00	1.80	2.52Low
60	5.80	1.00	7.70	4.30	5.20	2.80Low
61	8.30	1.10	9.80	3.50	1.30	3.67High
62	7.10	1.00	6.80	4.50	4.60	3.27Moderate
63	5.10	0.50	5.20	0.20	13.00	2.85High
64	7.40	0.40	7.50	2.80	5.90	3.26Moderate
65	5.90	1.70	7.00	3.70	5.70	2.80Low
66	6.90	2.50	7.50	5.10	2.00	3.06Moderate
67	8.30	0.70	5.40	3.90	5.70	3.26High
68	9.70	2.30	6.90	3.30	1.80	3.38High
69	7.30	2.20	9.70	1.90	2.90	3.15Moderate
70	5.40	1.30	9.20	0.10	8.00	2.69Low
71	9.10	1.10	5.60	4.20	4.00	3.54High
72	8.70	3.20	6.40	1.00	4.70	3.37High
73	8.80	3.20	10.00	0.80	1.20	3.58High
74	6.70	3.70	9.30	1.80	2.50	2.89Moderate
75	8.80	3.00	5.50	5.40	1.30	3.58High
76	6.60	3.60	6.90	0.10	6.80	2.99Moderate
77	9.50	0.40	6.60	5.70	1.80	3.44High
78	8.20	1.80	6.50	2.00	5.50	3.35High
79	8.40	3.00	9.00	2.80	0.80	3.26High
80	5.30	2.20	7.20	5.30	4.00	2.62Low
81	6.80	0.50	5.70	4.60	6.40	2.86High
82	5.40	2.80	5.40	4.90	5.50	2.86High
83	8.50	0.30	5.40	5.90	3.90	2.99High
84	9.10	3.80	9.90	0.90	0.30	3.08High
85	6.90	0.30	8.90	3.40	4.50	3.21Moderate
86	5.60	2.00	5.10	2.80	8.50	2.81High
87	5.30	0.50	5.60	3.90	8.70	2.90High
88	9.80	1.50	6.40	5.20	1.10	3.78High
89	6.10	3.90	5.10	5.80	3.10	2.96High
90	5.40	2.20	9.80	3.10	3.50	2.81Low
91	8.10	2.80	7.30	3.60	2.20	3.39High
92	5.20	1.10	9.80	5.30	2.60	2.82Low
93	7.30	2.50	6.40	1.10	6.70	2.67Moderate
94	7.90	0.30	9.90	5.80	0.10	3.08Moderate
95	8.50	2.10	6.50	4.90	2.00	3.10High
96	9.60	3.30	9.70	1.00	0.40	3.54High
97	8.10	1.70	9.70	3.90	0.60	3.45High
98	9.90	0.60	8.00	2.10	3.40	3.39High
99	9.80	3.40	9.20	0.70	0.90	3.15High
100	5.30	3.50	9.10	6.00	0.10	2.68Low

The Dataset

02 Data Cleaning & Preparation

After defining and discovering the data and what each column represents, we began the process of cleaning and arranging the data and making it ready for analysis.

- Dealing with missing values: There were no missing values in the data because it was complete.
- Dealing with outlier data: was good and there was no outlier value in salaries. This indicates that the data is integrated.

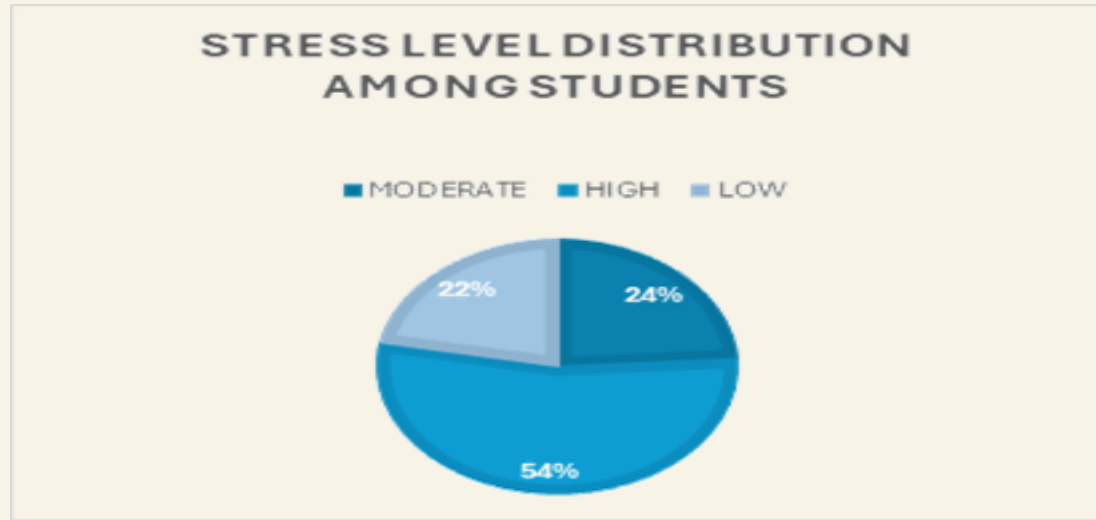
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Data visualization

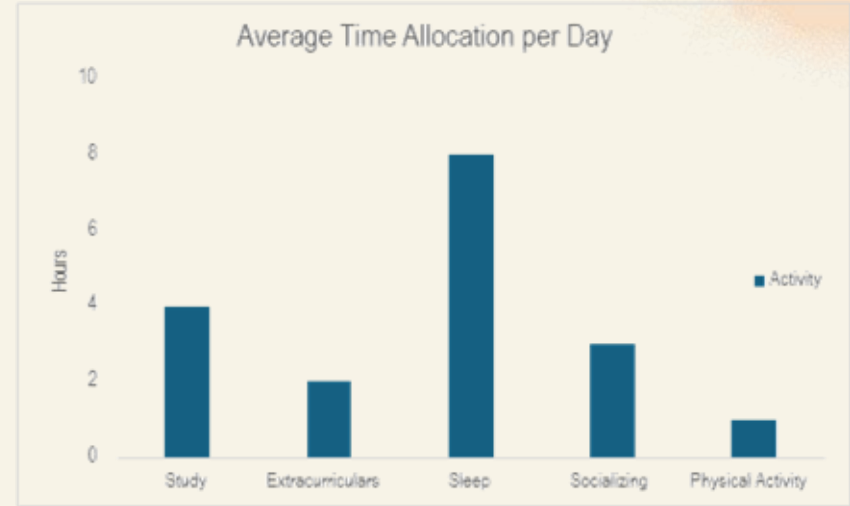
Pie Chart

BY using =COUNTIF() so, this chart shows the percentage distribution of students across different stress levels (Low, Moderate, High). So, we conclude that: a significant proportion of students fall into the "Moderate" stress category, followed by "Low" and "High."



Bar Chart

This chart compares the average daily time spent on various activities like studying, extracurriculars, sleeping, socializing, and physical activity.

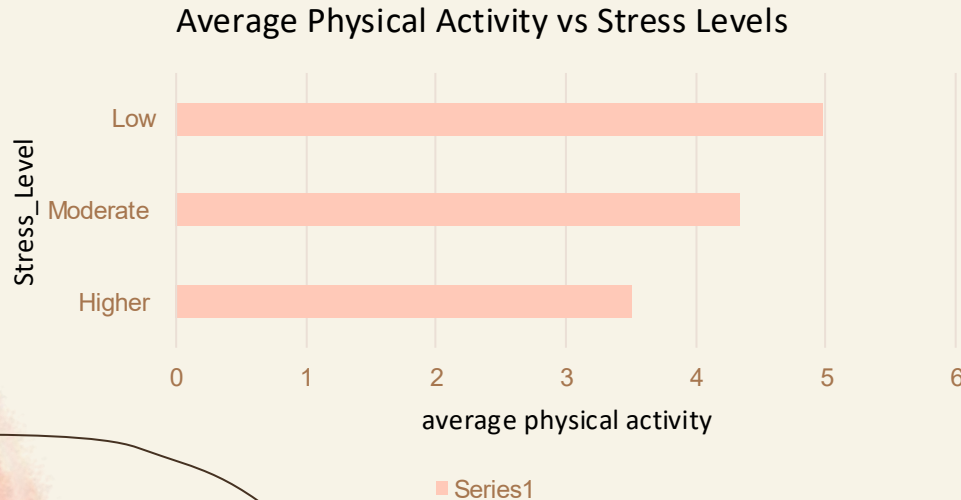


So, Students spend the most time sleeping, followed by studying. Physical activities and socializing take up less time on average.

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Bar Chart

This chart shows the relationship between stress level and physical activities.



Scatter Chart

This scatter chart explores the relationship between study hours and GPA.

Observations: A positive trend is visible: students who study more tend to have higher GPAs.



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Statistical Analysis

How significantly do factors like study hours, sleep hours, and stress levels impact a student's academic performance?

Predictive modeling techniques are used to analyze data and forecast outcomes by identifying relationships between variables. One commonly used technique is linear regression, which helps predict a continuous variable based on one or more predictors. In this case, linear regression can help us understand the impact of factors like study hours, stress levels, or sleep hours on GPA. By analyzing the relationship between these variables, we can quantify how changes in one factor influence GPA, determine which factors have the most significant effect, and identify patterns that can guide decision-making. This enables us to make data-driven conclusions and develop strategies to improve academic performance.

Correlation

Ranges of correlation

Range	Meaning
0 - 0.19	Very weak positive correlation
0.20 - 0.39	Weak positive correlation
0.40 – 0.59	Moderate positive correlation
0.60 - 0.79	Strong positive correlation
0.80 - 1	Very strong positive correlation
0- (-0.19)	Very weak negative correlation
(-0.19) – (- 0.39)	weak negative correlation
(-0.40) - (- 0.59)	Moderate negative correlation
(- 0.60) – (-0.79)	strong negative correlation
(-0.8) – (1)	Very strong negative correlation

Correlation		
Study hours	0.81	Very strong positive
Sleep hours	-0.04	Very weak negative
Stress level	0.67	Strong positive

Regression

Regression Statistics	
Multiple R	0.8091875
R Square	0.6547845
Adjusted R Square	0.6512619
Standard Error	0.1877737
Observations	100

Strong positive relationship

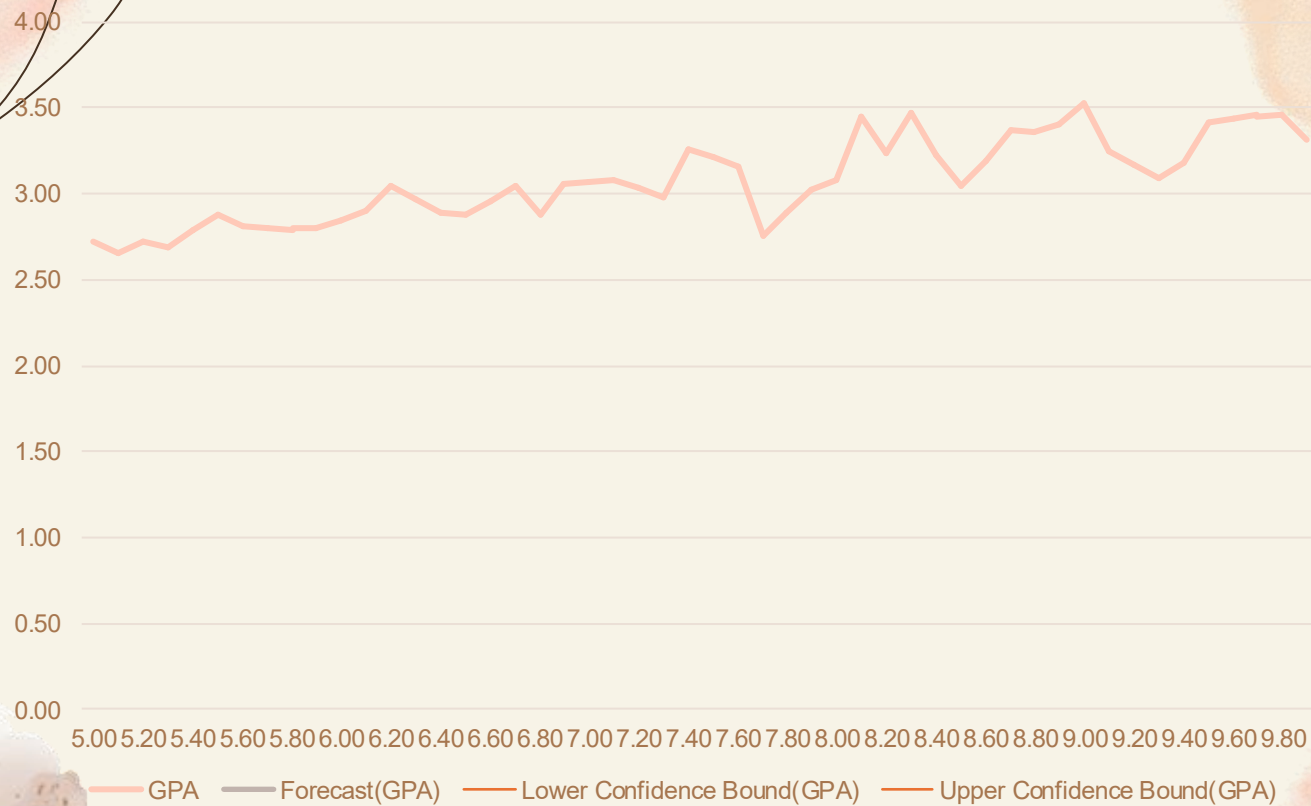
Regression

ANOVA					
	df	SS	MS	F	Significance F
Regression	1	6.55395979	6.55395979	185.8806285	2.29792E-24
Residual	98	3.45537921	0.035258972		
Total	99	10.009339			

Regression

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	1.9046866	0.088807761	21.44729936	8.54573E-39	1.72845053	2.080922754	1.72845053	2.080922754
Study_Hours_Per_Day	0.159275	0.01168236	13.63380462	2.29792E-24	0.136091751	0.182458278	0.136091751	0.182458278

- This p-value indicates that the predictor “Study Hours Per Day” has a highly significant impact on GPA.



The chart is useful for predicting GPA based on study hours per day. Here's how:

Forecasting:

The graph plots the historical data to identify the trend between the Study Hours Per Day (X-axis) and the corresponding GPA (Y-axis). By analyzing these trends, it helps in predicting the GPA for new study hours.

2. Confidence Bounds:

The chart includes Lower Confidence Bound and Upper Confidence Bound, which give a range within which the predicted GPA is likely to fall. This adds reliability to the forecast by accounting for uncertainty.

3. Trend Analysis:

The line chart shows visually that as study hours increase, the GPA tends to rise gradually. This can also allow the user to estimate what the expected GPA would be for a certain number of study hours.

This chart will be useful in making predictions and decisions on how increasing study hours will improve one's GPA.

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Insights & Reporting

Key findings

1. Stress Level Patterns (Pie chart)

"High" stress levels are most common among students, potentially due to imbalanced schedules (e.g., too little sleep or excessive extracurricular commitments).

2. Impact of Physical Activity (bar chart)

A. Students spend the most time sleeping, followed by studying. Physical activities and socializing take up less time on average.

B. Moderate levels of physical activity correlate with lower stress levels. However, excessive activity (outliers) might negatively affect time allocation for academics.

3. Correlation Between Study Hours and GPA (Scatter plot)

Students with higher study hours per day tend to have better GPAs, suggesting a positive relationship between academic effort and performance.

Business Insights & Recommendations

○ Business Insights:

1. Time Allocation: Students should aim for a balanced schedule emphasizing study and moderate physical activity while ensuring sufficient sleep to optimize performance.
2. Stress Management: High stress levels may indicate the need for interventions, such as workshops on time management or stress-relief programs.

○ Recommendations:

1. Encourage a Balanced Lifestyle: promote initiatives that balance study, physical activity, and relaxation to improve overall performance and well-being.
2. Offer Support Services: introduce stress-management resources to address the high prevalence of stress among students.
3. Monitor Outliers: identify and support students with extreme behaviors (e.g., too much physical activity or insufficient study time) to ensure better academic outcomes.

The background features soft, abstract watercolor washes in shades of pink, peach, and light orange, primarily located in the corners. Thin, elegant black curved lines sweep across the page, framing the central text.

Thanks!